📘 REGULARIZATION – ALL TERMS YOU ARE LEARNING

Model

👉 A Python object that learns from data and makes decisions.

Example:

model = LogisticRegression()

Data

👉 The numbers you give to the computer so it can learn.

Example:

X = [[20], [30], [40]]
y = [0, 1, 1]

Feature

👉 One input value.

Example:

- Age is a feature

Target / Label

👉 The answer you want the computer to learn.

Example:

- 0 = No
- 1 = Yes

Logistic Regression

👉 A method that predicts **Yes or No** using numbers.

Prediction

👉 The answer the model gives.

Example:

0 or 1

Weight / Coefficient (`coef_`)

👉 A number that shows **how important a feature is**.

- Big number → very important
- Small number → not important

Bias / Intercept (`intercept_`)

👉 A starting value added to every decision.

Overfitting

👉 When the model **memorizes** instead of learning.

Regularization

👉 A rule that **stops the model from becoming too confident**.

Penalty

👉 The type of rule used to control confidence.

Types:

- L1
- L2

L2 Regularization

👉 Makes all weights **smaller but keeps them**.

L1 Regularization

👉 Makes useless weights **exactly zero**.

C (Regularization Strength)

👉 How strict the rule is.

- Small C → very strict
- Large C → less strict

Solver

👉 The method used to find the best weights.

Example:

solver='liblinear'

Fit

👉 The process of learning from data.

model.fit(X, y)

Train

👉 Same meaning as fit (learning step).

Predict

👉 Ask the model for an answer.

model.predict(X)

Coefficient Shrinking

👉 Weights becoming smaller due to regularization.

Feature Selection

👉 Removing useless features (done by L1).

🧠 ONE-SCREEN SUMMARY

Data → Model → Learns (fit)
Learns weights (coef_)
Regularization controls weights
L2 shrinks
L1 shrinks + removes
C controls strictness
